Chathuranga Weeraddana

Senior Research Fellow (Docent), University of Oulu, Finland
+358 449775219 • chathurw@gmail.com • LinkedIn
Google Scholar • Homepage

1. Objectives

My work spans **wireless communications, signal processing, machine learning, and optimization**, with a focus on the **theoretical and algorithmic foundations of intelligent future communication systems**. My research has addressed radio resource management, distributed optimization, learning, and network intelligence across multiple generations of wireless systems. I am particularly interested in translating rigorous technical research into **high-impact 5G and 6G innovations**, with emphasis on their practical relevance to cellular system design, standardization, and intellectual property.

2. Research Interests

Future wireless and 6G systems, radio resource management and optimization, signal processing, distributed optimization, distributed machine learning, multi-agent systems, AI-native communication networks, and intelligent cellular system design.

3. Teaching Interests

Including, but not limited to: Mathematical optimization, machine learning, statistical signal processing.

4. Degrees

University of Oulu **Finland**
Doctor of Science, Telecommunication Engineering *2011*
Thesis: Optimization Techniques for Radio Resource Management in Wireless Communication Networks
Reviewers: Prof. Leandros Tassiulas (Yale University, USA), Prof. Lars K. Rasmussen (KTH, Sweden)
Opponents: Prof. Lars K. Rasmussen (KTH, Sweden), Prof. Olav Tirkkonen (Aalto University, Finland)

Asian Institute of Technology **Thailand**
Master of Engineering, Telecommunication Engineering *2007*
Thesis: Performance Analysis of OFDM Systems with Random Residual Frequency Offset

University of Moratuwa **Sri Lanka**
Bachelor of Science, Electronic and Telecommunication Engineering *2004*

5. Leadership Experiences

- **DESIRE6G** (*Deep Programmability and Secure Distributed Intelligence for Real-Time End-to-End 6G Networks*), 2023-2025, funded under Horizon Europe / SNS JU
  - Managed the University of Oulu's contributions to the Horizon Europe DESIRE6G, coordinating technical activities, deliverables, and collaboration with consortium partners.
- **IEEE MERCon**, the flagship international conference of the University of Moratuwa, Sri Lanka
  - International Relations Chair, 2022
  - Publication Chair, 2020
- **NCTM, National Conference on Technology and Management**, organized by the Research Center, SLIIT, Sri Lanka
  - Co-Chair, 2016

6. Research and Professional Experience

CWC-RT, University of Oulu — **Finland**
Docent — *2024/09 - To Date*
Working in the subgroup Intelligent Connectivity and Networks/Systems Group (ICON)

CWC-RT, University of Oulu — **Finland**
Senior Research Fellow — *2022/08 - 2024/09*

Dept. of Electronic & Telecommun. Engineering, University of Moratuwa — **Sri Lanka**
Senior Lecturer (equivalent to Associate Prof.) — *2018 - 2022/07*

Dept. of Electrical & Computer Engineering, SLIIT — **Sri Lanka**
Senior Lecturer (equivalent to Associate Prof.) — *2015 - 2018*

Dept. of Automatic Control, KTH Royal Institute of Technology — **Sweden**
ACCESS Postdoc Researcher — *2012-2014*

7. Selected Publications

1. H. Abeynanda, **P. C. Weeraddana**, and C. Fischione, "On the Characteristics of the Conjugate Function Enabling Effective Dual Decomposition Methods," *IEEE Transactions on Signal Processing*, vol. 74, pp. 572-588, Jan. 2026.
2. T. AlShammari, **P. C. Weeraddana**, and M. Bennis, "BayGO: Decentralized Bayesian Learning and Information-Aware Graph Optimization Framework," *IEEE Transactions on Signal Processing*, vol. 72, pp. 2101-2116, Apr. 2024.
3. S. Ranaweera, **P. C. Weeraddana**, P. Dharmawansa, and C. Fischione, "On the Convergence of Inexact Gradient Descent With Controlled Synchronization Steps," *IEEE Signal Processing Letters*, vol. 30, pp. 703-707, May. 2023.
4. H. Abeynanda, **P. C. Weeraddana**, G. H. J. Lanel, and C. Fischione, "On the Primal Feasibility in Dual Decomposition Methods Under Additive and Bounded Errors," *IEEE Transactions on Signal Processing*, vol. 71, pp. 655-669, Feb. 2023.
5. M. Ding, S. Atapattu, **P. C. Weeraddana**, and J. Evans, "Energy-Efficiency Fairness of Interference Multi-Relay Networks for Multi-User Communication," *IEEE Communication Letters*, vol. 25, no. 3, Mar. 2021.
6. S. Magnússon, **P. C. Weeraddana**, M. Rabbat, and C. Fischione, "On the convergence of alternating direction lagrangian methods for nonconvex structured optimization problems," *IEEE Transactions on Control of Network Systems*, vol. 3, no. 3, pp. 296–309, Oct. 2016.
7. S. Magnússon, **P. C. Weeraddana**, and C. Fischione, "A distributed approach for the optimal power flow problem based on admm and sequential convex approximations," *IEEE Transactions on Control of Network Systems*, vol. 2, no. 3, pp. 238–253, Oct. 2015.
8. G. Athanasiou, **P. C. Weeraddana**, C. Fischione, and L. Tassiulas, "Optimizing client association for load balancing and fairness in millimeter-wave wireless networks," *IEEE/ACM Transactions on Networking*, vol. 23, no. 3, pp. 836-850, Mar. 2014.
9. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, A. Ephremides, and Carlo Fischione, "Weighted sum-rate maximization in wireless networks: A review," *Foundations and Trends in Networking, Now Publishers Inc.*, vol. 6, no. 1-2, pp. 1–163, 2012.
10. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, and A. Ephremides, "Weighted sum-rate maximization for a set of interfering links via branch and bound," *IEEE Transactions on Signal Processing*, vol. 59, no. 8, pp. 3977–3996, Aug. 2011.

11. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, and A. Ephremides, "Multicell downlink weighted sum-rate maximization: A distributed approach," *IEEE Transactions Signal Processing*, vol. 61, no. 3, pp. 556–570, February 2013.

8. Teaching Experience

Div. of Network and Systems Engineering, KTH Royal Institute of Technology **Sweden**
Lecturer, Graduate Course *2022/03-2022/06*
MM Optimization Algorithms with Machine Learning Applications
`https://chathugithub1.github.io/Home/courses/courseinfo.html`

Faculty of Engineering, University of Moratuwa **Sri Lanka**
Lecturer, Graduate Cousrses *2018-2022*
Engineering Decision Theory, Broadband Communications, Digital Communications, Advanced Instrumentation

Faculty of Engineering, University of Moratuwa **Sri Lanka**
Lecturer, Undergraduate Courses *2018-2022*
Introduction to Telecommunication, Teletraffic Engineering

Dept. of Electrical & Electronics Engineering, University of Peradeniya **Sri Lanka**
Lecturer, Graduate Course *2018- 2022*
Optimization in Communication Systems

Dept. of Electronic Engineering, Yangon Technical University **Myanmar**
Lecturer, Graduate Course *2018 - 2019*
Introduction to Mathematical Optimization

Dept. of Electrical & Computer Engineering, SLIIT **Sri Lanka**
Lecturer, Graduate Courses *2015 - 2018*
Several Communication and Signal Processing Courses

Dept. of Electrical & Computer Engineering, Curtin University **Australia**
Lecturer, Undergraduate Courses through SLIIT *2015 - 2018*
Several Communication and Signal Processing Courses

Dept. of Automatic Control, KTH Royal Institute of Technology **Sweden**
Lecturer, Graduate Courses *2013-2014*
Network Optimization, Principles of Wireless Sensor Networks

Centre for Wireless Communications, University of Oulu **Finland**
Lecturer, Graduate Courses *2010*
Convex Optimization with Applications in Wireless Communications and Networks

9. Invited Talks and Visits for Research/Teaching

KTH Royal Institute of Technology **Sweden**
Guest Researchers *2022/03/01 - 2022/06/01*

Dept. of Electrical & Electronics Engineering, University of Peradeniya **Sri Lanka**
Visiting Lecturer *2018-2022*

Dept. of Electronic Engineering, Yangon Technical University **Myanmar**
Visiting Lecturer *2019/08/19 - 2019/09/01*

Dept. of Electronic Engineering, Yangon Technical University **Myanmar**
Visiting Lecturer *2018/09/10 - 2018/09/20*

16th International Conference on Industrial and Information Systems (IEEE ICIIS) **Sri Lanka**
Guest Speaker *2021*
Session: 5G Evolution and 6G

Faculty of Graduate Studies, University of Moratuwa **Sri Lanka**
Resource Person *2022/08/23*
Event: Research Lounge

10. Awards and Honours

- **ACCESS Post-Doctoral Fellowship:** Among 1 of the 3 post-doctoral fellows selected from ACCESS Linnaeus Centre, KTH Royal Institute of Technology, Sweden, 2012
- **Best doctoral thesis award:** Dept. of Comm. Eng., University of Oulu, Finland, 2011
- **Second best paper award:** Asilomar Conference on Signals, Systems and Circuits 2010
- **GETA grant:** Graduate School in Electronics, Telecommunications and Automation grant for a 4 years of graduate school position in University of Oulu, Finland, 2011
- **Students' research grants:** Tauno Tönning Säätiö, Finland, 2011, 2009
- **Students' research grants:** Oulun Yliopiston Tuksäätiö, Finland, 2010
- **Students' research grants:** Nokia Foundation, Finland, 2010, 2009
- **Students' research grants:** Tekniikan Edistämissäätiö Stiftelsen, Finland, 2008
- **The AIT alumni association prize:** in recognition of the most outstanding academic performance among students in the School of Engineering and Technology, AIT, Thailand, 2007
- **The Sonera prize:** in recognition of the most outstanding academic performance among students in Telecommunication, AIT, Thailand, 2007
- **King's Scholarship:** from His Majesty King Bhumibol Adulyadej of the kingdom of Thailand, 2005

11. Other Key Academic Merits

- **Exemplary Reviewer:** Certificate of appreciation from IEEE Communication Society, IEEE Wireless Communications Letters, 2020,2021
- **Reviewer:** IEEE Transactions of Communications, IEEE Wireless Communications Letter, IEEE Transactions of Signal Processing, and IEEE Transactions of Wireless Communications, among others
- Completed an Induction Training course at the Staff Development Centre, Open University of Sri Lanka (2019), covering key pedagogical frameworks including outcomes-based education (OBE), constructive alignment, scaffolding, constructivism, curriculum design and mapping, program design, and assessment methods.

Referees

Professor Carlo Fischione
Department of Network Systems Engineering
KTH, The Royal Institute of Technology
Osquldas Väg 10,
Stockholm, 10044, Sweden
carlofi@kth.se
46 736 322561
46 879 07329

Professor Sindri Magnússon
Department of Computer and Systems Sciences
Stockholm University
Borgarfjordsgatan 12
SE-164 07 Kista, Sweden
sindri.magnusson@dsv.su.se
46 72 303 9104

Professor Marco Levorato
Computer Science Department
3206 Donald Bren Hall
University of California, Irvine
Irvine, CA, 92697-2800
levorato@uci.edu
1 949 824 2175